

Fig. 7: Class library in client-side

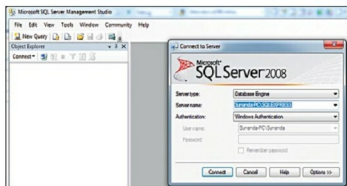


Fig. 8: SQL Server Management Studio

It is time to design the layout for our window. In line five of `MainWindow.xaml` code, we create `StackPanel` to define the layout, which will stack all elements inside it vertically (by default). Once the layout is defined, it is time to create the elements inside it. Since we want to create a tabbed UI, we start with creating `TabControl` of height 365 units. `TabControl` has two tabs: Registration and Employee Data.

Registration tab. Under Registration, we first define the layout. This time we use another type of layout, called `Grid`. `Grid` layout is more like a table in HTML, which can have multiple rows and columns. Here, we require 11 rows and two columns, which are specified by `RowDefinition` and `ColumnDefinition`, respectively. Then, we add different fields, `TextBlocks`, `TextBoxes`, `Expander`, a button, and different columns and row numbers inside `Grid`. Inside `Expander`, we have some more port-related fields, `TextBlocks` and `TextBoxes`, that

get populated once the application starts and the specified port number opens and the RFID tag is brought near the RFID reader. User-related details have to be filled in by the admin. Now, if all data is filled, click Register and after successful registration, "Registered !!" will appear in the UI.

Employee data.

Under this tab, like before, we first create the layout as `Grid` with `RowDefinition` and `ColumnDefinition`. Then, we create other fields, `TextBlocks`, `TextBoxes` and buttons. Here, the admin gives the employee ID as input in the textbox and clicks on Search and

then if a matching ID exists in the database, the result is displayed in the result `TextBlock`.

Since designing even a small layout involves a lot of code work, a step-by-step view of the layout design would be further explained, which is included in the EFY DVD. Please follow it in case you need a bird's-eye view of the layout-construction stage.

Employee attributes. Create the necessary data object, or rather model object, for employee attributes that are to be further used for processing in the client-side.

1. Create a class library and name is `AttendanceManagementLibrary`. Add a new project to the solution. Under New Project-> Visual C#, we find the option to create a class library as shown in Fig. 7.

2. Once the project gets added to our solution `AttendanceClientSide` in `Class1.cs` file, we add presentation data model class named `EmpAttendanceLib` with some employee proper-

ties, like `EmpId`, `Rfid`, `Name`, `Address`, `LoginTime` and `LogoutTime`. The code to achieve this is listed below.

```
public class EmpAttendanceLib
{
    public string EmpId { get; set; }
    public string Rfid { get; set; }
    public string Name { get; set; }
    public string Address { get; set; }
    public string LoginTime { get; set; }
    public string LogoutTime { get; set; }
}
```

Employee attributes are used to display these in the client UI and for processing various other logics.

With these steps completed, our client-side is almost ready to be attached to the server-side. After server logic is written, we will come back and write button-click logics for the client-side.

Database section

The next big task is to create the database. Before starting this, we should make sure that we already have SQL server 2008, `SQLManagementStudio_x64_ENU` and `SQLSERVER_x64_ENU` installed on our machine. Configuration of SQL Server Management Studio is shown in Fig. 8. (Installation and configuration details are included in EFY DVD in .pdf format.)

To create the database, we will follow database-first approach as our entity data model (EDM) designing approach.

In our database, we have two tables: `EmployeeDetail` and `CurrEmpDetail`. `EmployeeDetail` is for storing employee-related data, like names, addresses, RFID numbers and employee ID numbers (primary key). `CurrEmpDetail` contains details of log-in and log-out time, number of times the employee has logged-in/logged-out, along with ID of the employee.

Steps to create the database are given below:

1. To create the database, open another instance of Visual Studio in admin mode and create a new project of type Console Application.

Here, we name the solution `AttendanceServerSide` and the console